

Zilin Wang

Email: wzlwzlwzl@stu2023.jnu.edu.cn Homepage: <https://zilamwong2005-math.github.io/>

EDUCATION

Jinan University, Guangzhou, China

Sep. 2023 – Jul. 2027 (Expected)

B.Sc. in Mathematics

Total Average: 84/100

Main Courses: Abstract Algebra, Complex Analysis, Real Analysis(95), Functional Analysis, Numerical Analysis(89), Mathematics for Finance(96), Ordinary Differential Equations(89), Physics(90), Differential Geometry.

Research Interests: Mathematical Physics, Dynamical Systems, Applied Mathematics.

PUBLICATIONS AND MANUSCRIPTS

Z.Wang, Q.Liu, Z.Chen, and S.Fan., "Lipschitz-Regularized Path-Strength Refactoring for Differentiable Architecture Search," *Neurocomputing*, Major Revision, 2026. [Available at SSRN]

ADDITIONAL MATH EXPERIENCE

Undergraduate Training Program for Innovation and Entrepreneurship | Project Leader

May. 2025 – Apr. 2026

- Decoupled the traditionally unstable bi-level optimization into a sequential single-level framework, effectively suppressing the "performance collapse".
- Introduced **Lipschitz constraints** to reformulate the neural search space, providing a rigorous connection between loss landscapes and path gradients through the principles of Real Analysis.
- Refactored architecture selection as a fitness evaluation task via **path-gradient integrals**.
- Authored the paper *Lipschitz-Regularized Path-Strength Refactoring for Differentiable Architecture Search*.

Incoming Research Internship | University of Alberta, Canada

Jul.2026 – Sep.2026 (Expected)

- Focus on Stochastic Dynamics and its applications under supervision of Prof.Zhongwei Shen.

WORK

Associate | Publicity and Media Center, School of Mathematics

2023-2024 Academic Year

Director | Publicity and Media Center, School of Mathematics

2024-2025 Academic Year

HONORS AND AWARDS

Outstanding Student Cadre Scholarship

2023–2024 Academic Year

Academic Excellence Scholarship (Third-class)

2024–2025 Academic Year

China Undergraduate Mathematical Contest in Modeling, Provincial Third Prize

Oct. 2024

Asia and Pacific Mathematical Contest in Modeling (APMCM), National Second Prize

Nov. 2024

The Chinese Mathematics Competitions (Mathematics Major Group), Provincial Third Prize

Nov. 2025